

## Stroke and Post-stroke rehabilitation

An **ischemic stroke** occurs when blood flow to a part of the brain is blocked, usually due to a blood clot or a narrowing of the arteries. This lack of blood supply deprives brain tissue of oxygen and nutrients, leading to cell damage or death. Ischemic strokes are the most common type of stroke, accounting for about 87% of all cases. Risk factors include high blood pressure, diabetes, smoking, and high cholesterol.

**Post-stroke symptoms** vary depending on the area of the brain affected and can include physical, cognitive, and emotional challenges. Common symptoms include weakness or paralysis on one side of the body, difficulty with speech and swallowing, impaired vision, balance problems, memory loss, and fatigue. Emotional changes, such as depression and anxiety, are also common.

**Post-stroke rehabilitation** aims to help survivors regain independence and improve their quality of life. It involves physical therapy to restore movement and strength, occupational therapy to relearn daily tasks, and speech therapy to address communication issues. Rehabilitation often incorporates exercises to improve mobility, coordination, and cognitive skills, and assistive technologies or intelligent coaching systems may be used to enhance recovery.

### Persona 1: Eleanor

- **Age:** 72
- **Occupation:** Retired Librarian
- **Stroke Type:** Ischemic stroke affecting the right side
- **Mobility Status:** Hemiparesis (weakness) on the left side, uses a cane for walking.
- **Cognitive Status:** Mild cognitive impairment (slowed processing, occasional memory lapses).
- **Goals:**
  - Improve left-side mobility and hand function for daily activities like cooking and gardening.
  - Maintain independence and reduce the risk of falls.
  - Stay socially engaged to prevent isolation.

### Needs:

- **Rehabilitation Focus:** Motor recovery, balance improvement, and cognitive stimulation.
- **Technology Comfort Level:** Low to medium; familiar with using smartphones but hesitant about new technology.

- **Support System:** Family (children and grandchildren) is supportive, but she values her independence.

### ***Rehabilitation Strategy:***

- **Physical Therapy:** Focus on exercises to strengthen the left leg and arm, along with balance training.
- **Occupational Therapy:** Fine motor skills rehab with everyday tasks like buttoning clothes and holding utensils.
- **Cognitive Exercises:** Memory games or brain exercises that can be done in short sessions.

## **Persona 2: Carlos**

- **Age:** 50
- **Occupation:** Software Engineer
- **Stroke Type:** Ischemic stroke affecting left-side motor control and speech (aphasia).
- **Mobility Status:** Weakness in the right arm and leg, difficulty with fine motor tasks, such as typing.
- **Cognitive Status:** Mild language processing difficulties, but cognitively sharp otherwise.
- **Goals:**
  - Restore fine motor skills to return to work (typing, coding).
  - Improve speech clarity and reduce frustration with communication.
  - Regain independence in daily activities, including driving.

### ***Needs:***

- **Rehabilitation Focus:** Upper-body motor recovery, speech therapy, and emotional support.

- **Technology Comfort Level:** High; comfortable with using apps, wearables, and online resources.
- **Support System:** Strong family support (spouse and two kids) but feels pressure to resume work and provide.

### *Rehabilitation Strategy:*

- **Motor Skill Rehab:** Customized hand exercises (using tech tools) to regain fine motor control and dexterity, especially for typing.
- **Speech Therapy:** Intensive speech therapy with speech recognition apps to improve communication at work and home.

### **Persona 3: Lily**

- **Age:** 25
- **Occupation:** student
- **Stroke Type:** Ischemic stroke affecting the left side, leading to right-side hemiplegia (paralysis).
- **Mobility Status:** Uses a wheelchair for longer distances but is learning to walk with assistance. Limited use of the right arm.
- **Cognitive Status:** No significant cognitive deficits but struggles with frustration and adapting to school after the stroke.
- **Goals:**
  - Regain as much use of her right arm and leg as possible.
  - Return to university and participate in activities with friends.
  - Reduce frustration and build confidence in her recovery.

### ***Needs:***

- **Rehabilitation Focus:** Mobility recovery, integration of therapy into university life, and emotional support.
- **Technology Comfort Level:** Comfortable using tablets and enjoys gamified learning experiences.
- **Support System:** Strong support from parents and university but feels disconnected from peers because of her physical limitations.

### ***Rehabilitation Strategy:***

- **Physical Therapy:** Fun, game-based exercises that help her engage with therapy and improve right-side mobility, such as using interactive balance and strength exercises.
- **University Integration:** Work with teachers to provide a supportive learning environment (e.g., adaptive tools for writing, extra time for physical activities).